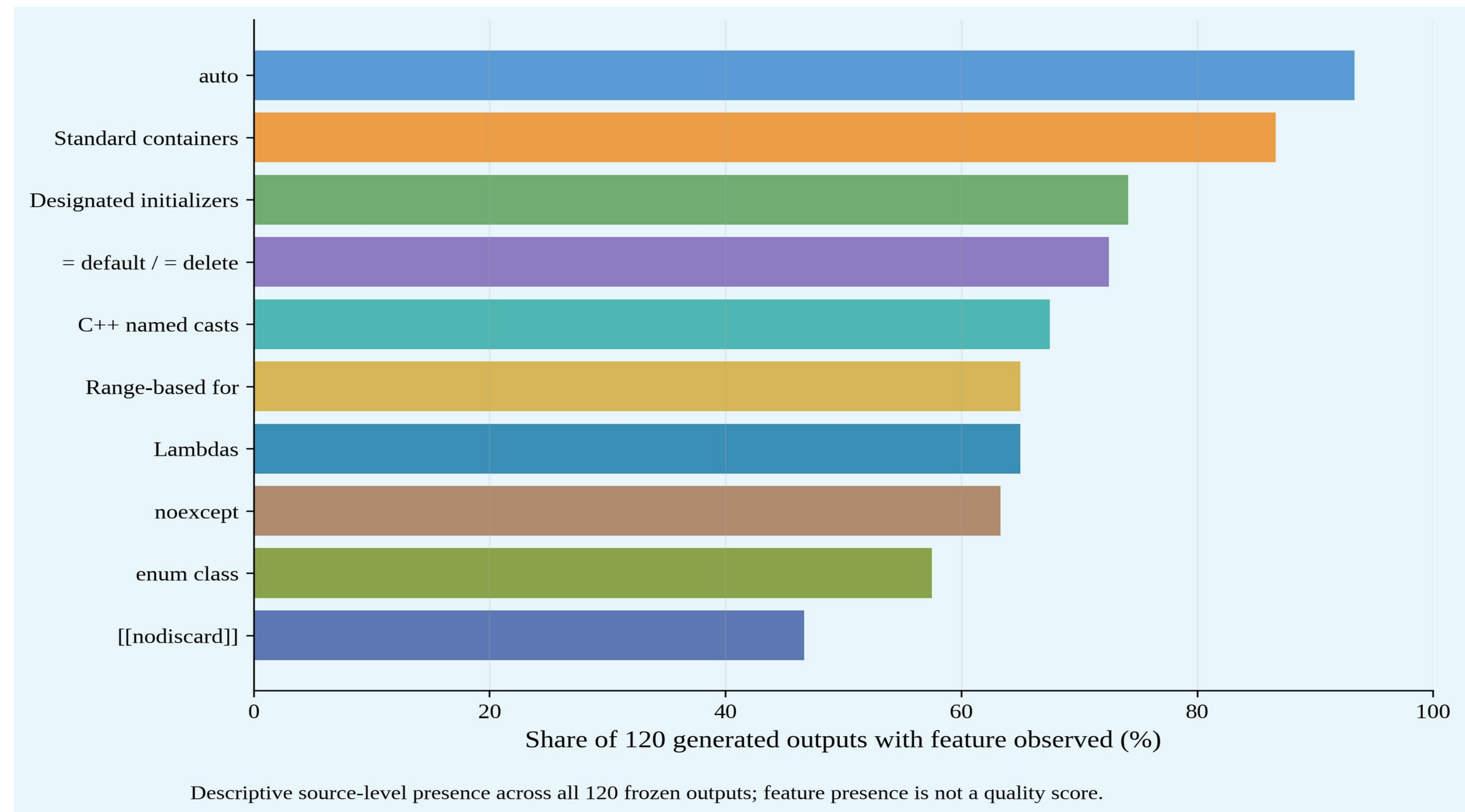


Frequently Observed Modern C++ Features and Practices

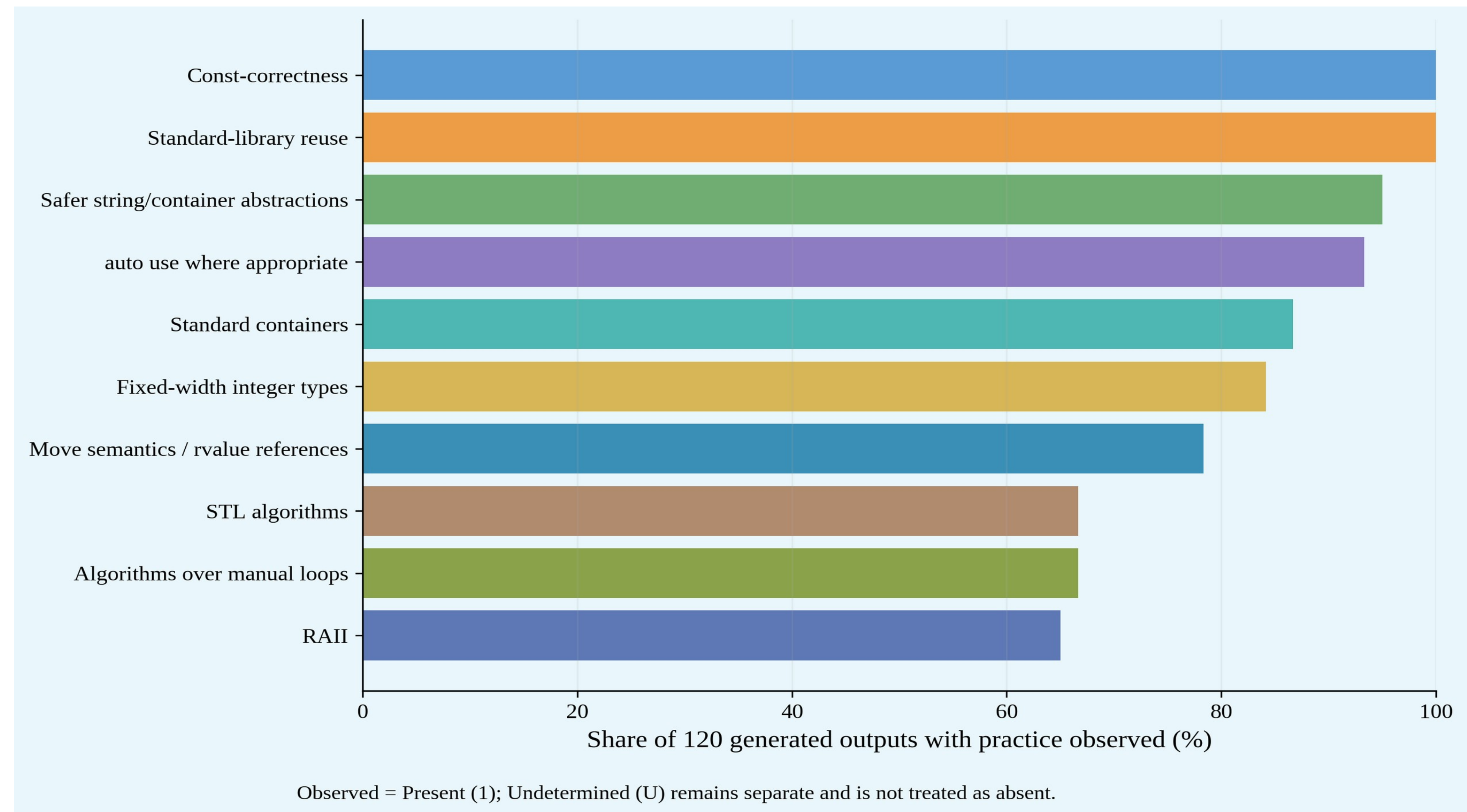
Source-level descriptive prevalence across the 120 frozen LLM-generated C++ outputs

Figure 5. Most Frequently Observed Concrete Modern C++ Features



This directly answers which concrete language/library features were most frequently observed in the generated source; it is descriptive, not a quality score.

Figure 6. Most Frequently Observed Broader Modern C++ Practices



Semantically duplicated inventory labels were consolidated for display readability; Undetermined cases are retained separately and not counted as absent.

Modern C++ Feature Wall — Examples from the 84 Concrete Features Evaluated

1. Memory Safety

std::unique_ptr • std::shared_ptr • std::weak_ptr • std::make_unique • std::make_shared • RAII

2. Type Safety

enum class • static_cast • dynamic_cast • const_cast • concepts • explicit

3. Bounds & Data Safety

std::span • gsl::span • std::array • std::vector::at • std::string_view • std::byte

4. Compile-Time Enforcement

constexpr • constexpr • constexpr • static_assert • requires • if constexpr

5. Modern Language Feature Adoption

auto • structured bindings • designated initializers • inline variables • fold expressions • CTAD

6. Standard Library & Generic Programming

std::vector • std::map • std::set • std::unordered_map • std::ranges • std::optional

7. Modern Programming Style

lambdas • range-based for • auto • const • noexcept • override

8. Concurrency & Parallelism

std::thread • std::jthread • std::async • std::atomic • std::mutex • std::lock_guard

9. Error Handling & Robustness

try/catch • std::optional • std::expected • std::error_code • noexcept • [[nodiscard]]

10. Platform & Systems Awareness

alignas • std::byte • std::endian • std::bit_cast • std::span • std::atomic

11. Code Quality & Maintainability

[[nodiscard]] • [[maybe_unused]] • [[deprecated]] • override • final • = default / = delete